

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE Spring-2026, Midterm Exam

Title: Social, Professional & Ethical Issues in Computing

Course Code: CSE-4805, Full Marks: 30, Time: 1 Hours 30 Minutes

			Mark	CLO
1	a)	Data centers now consume electricity at the scale of small nations, largely due to AI workloads. Is the hardware community morally obligated to prioritize energy efficiency over raw performance? Who should bear this responsibility: chip designers, cloud providers, AI developers or anyone else?	5	CLO1
	b)	Describe the impact of information technology on the following sectors: (i) Business (ii) Medicine	5	CLO3
2	a)	Your smartphone knows your location, sleep pattern, purchase habits, even your emotional state through app usage. You agreed to these <i>terms of service</i> that nobody reads. Is "informed consent" still a meaningful concept when the average app's privacy policy takes 45 minutes to read fully? Who is really at fault: the user who clicks "agree" or the system designed to exploit that click? OR When Edward Snowden revealed in 2013 that NSA was secretly mass-surveilling millions of ordinary citizens, the world was outraged, governments were questioned, protests erupted, and it dominated global headlines for months. Yet today, people willingly hand over their location, face, voice, relationships, and daily habits to private corporations through apps they chose to install, and almost nobody protests. The breach is arguably larger, more personal, and more continuous than anything Snowden exposed. So why the silence?	5	CLO1
	b)	Facial recognition is now embedded in airports, shopping malls, and even school campuses across several countries. A person can be identified, tracked, and profiled without ever giving consent or even knowing it happened. Does the convenience and security benefit justify this level of passive surveillance, or does it fundamentally violate the right to anonymous existence in public space?	5	CLO3
3	a)	Being anonymous online can be both a blessing and a curse. It helps whistleblowers expose corruption, protects abuse victims from being identified, and allows people living under oppressive governments to speak freely. But that same anonymity also lets cyberbullies harass people without consequences, and allows extremists to spread hate without being caught. Given that the same anonymity protects both heroes and villains; what moral framework would you suggest to handle this?	5	CLO1
	b)	The Epstein files exposed some of the most powerful people in the world with real evidence, real victims, and real documentation. Yet nothing happened. No major arrests, no accountability, the conversation just faded away. But in Bangladesh, an ordinary person writes a Facebook post criticizing a politician and gets arrested within 48 hours under the Digital Security Act. Same world, same technology, completely different justice. Why is the law lightning fast against the powerless but suddenly blind when the powerful are involved? Is this a broken system, or a system that was never meant to treat everyone equally in the first place? OR Why was Section 57 of the ICT Act 2006 considered controversial? What were its major consequences for freedom of expression, and how was it addressed through later legislative reforms?	5	CLO2

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. Engineering in CSE

Midterm Examination, Spring 2026

Course Code: MGT-3601

Course Title: Industrial Management

Time: 1 hour 30 minutes

Full Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions	
CLO1	Explain the theories and principles of modern management and apply the concepts to the management of organizations in private and public sector
CLO2	Understand how managers can effectively plan in today's dynamic environment,
CLO3	Identify what strategies organizations might use to become more innovative and explain how the industrial company markets and price its products and also how the company deal with its environment.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	What is an organization? What are the four basic resources that organizations use for its formation? Identify these four resources of IIUC.	CLO1	U	5
1)	b)	How is a 'Manager' defined? What is the role of top managers?	CLO1	An	5
2)	a)	What is the primary role of a corporate board of directors in shaping the organizational internal environment?	CLO2	R	5
2)	b)	Discuss the importance and determinants of an organization's culture and how the culture can be managed?	CLO2	Ap	5
3)	a)	What are five core dimensions of the job characteristics approach? Discuss these dimensions in designing jobs.	CLO3	U	5
3)	b)	How would you use the basic alternative approaches to design jobs?	CLO3	E	5
Or					
3)	a)	What are the drawbacks of a tall organizational structure.	CLO3	U	5
3)	b)	Describe basic ways in which positions within an organization can be differentiated.	CLO3	E	5

1. a) A mobile banking app requires login using OTP and role-based access (admin vs user). Differentiate between authentication and access control. Explain how both work together in this system. CO1 An 5
- Or,
- 1 a) A company deploys a system where employees must enter a password and approve a push notification on their registered phone. An attacker steals a password but cannot log in. Which specific security mechanism(s) from the X.800 list are being used? Justify your answer. CO1 An 5
1. b) Differentiate between active and passive attacks? If an attacker intercepts a fund transfer request in an online banking system and replays it multiple times to steal money – what type of attack is this? Justify your answer. CO2 E 5
2. a) Determine the ciphertext for the message “meet me after the toga party” using the following techniques:
i. Caesar cipher(key = 5)
ii. Single columnar transposition cipher(key= 42135)
Also, briefly provide cryptanalysis of each method. CO1 U 5
2. b) Differentiate between stream and block cipher? “The set of rational numbers is a field”- justify this statement with proper mathematical analysis. CO2 E 5
- Or,
- 2 b) What is prime field and extension field? Write down the polynomials elements in $GF(2^3)$. CO2 E 5
3. a) Zoom uses a KDC model where a central server generates a unique session key for each meeting, encrypted with each participant’s master key. What is the advantage of using a session key per meeting rather than a long-term shared key? What happens if the KDC is compromised? CO1 An 5
- b) Compare between link encryption and end to end encryption. CO1 U 5

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring 2026

Course Code: CSE 4871

Course Title: Neural Networks and fuzzy system

Total marks: 30 Time: 1 hour 30 minutes

[Figures in the right-hand margin indicate full marks.]

	M	CO
1.		
a) Why do we need Neural Networks when simple classifiers already exist?	2	1
b) A dataset contains two classes that can be separated by a straight line. Would you use a perceptron for this problem? Justify your answer.	3	2
OR		
b) A dataset contains two classes arranged like XOR. Why is perceptron not a suitable model here? What model would be better?	3	2
c) Suppose a model is not learning properly. Which among the following would you check first: weights, loss function, learning rate, or optimizer? Explain your reasoning.	5	2
2.		
a) "A neural network can work well even without an activation function." Do you agree? Justify your answer.	2	2
OR		
a) "Loss function is necessary in learning based algorithms," Do you agree? Justify your answer.	2	2
b) Suppose a neural network has the following structure: Input layer with 5 nodes Hidden layer 1 with 3 nodes Hidden layer 2 with 4 nodes Output layer with 3 nodes Calculate the total number of parameters in this network. Ignore the bias terms and show the detailed calculation.	2	1
c) Consider an XOR network with two hidden neurons. The initial weights are $w_{11} = w_{12} = w_{21} = w_{22} = 1$. The neurons use a step activation function with threshold, $\theta = 1$, where the output is 1 if $y_{in} \geq 1$, otherwise 0. Assume the learning rate, $\eta = 1.5$. For all XOR input combinations, answer the followings for one epoch only: i. Calculate the net input and output of each hidden neuron. ii. Show the weight update calculation for this learning rate. iii. Interpret the role of each hidden neuron conceptually.	2+ 2+ 2	2
3.		
a) How does backpropagation help reduce the overall loss of a neural network?	2	2
b) Is it necessary for the activation function of the output layer to be different from that of the hidden layers? Explain with an example.	5	2
OR		
b) Suppose a neural network has only one hidden layer. Can it solve a non-linear problem if a sufficient number of hidden neurons and a suitable activation function are used? Discuss.	5	2
c) Why is one-hot encoding necessary for ground truth labels in multi-class classification? Explain with reference to loss calculation.	3	2

International Islamic University Chittagong

Dept. of Computer Science & Engineering (CSE)

B.Sc. in CSE, Semester Mid-Term Examination, Spring 2026

Course Code: CSE 4875 Title: Pattern Recognition and image processing

Total Marks: 30

Time: 1.5 hours

[Answer all the following Questions. Figures in the right-hand margin indicate full marks]

CO DL

- 1(a) "One picture is worth more than ten thousand words" – Explain the specific fields of image processing that support this statement. 2 CO1 C2
- 1(b) Why are Sampling and Quantization considered the most critical steps in converting a continuous scene into a digital image? 2 CO1 C2
- 1(c) The continuum from image processing to computer vision involves low-, mid-, and high-level processes. Explain the necessity of each level with a real-world example. 3 CO1 C2
- 1(d) "A digital image is a representation of a two-dimensional image as a finite set of digital values." Justify this using a mathematical formula representing a discrete image function. 3 CO1 C2
- 2(a) Justify the statement: "Applying a Low-pass filter on an image results in a blurrier image." Explain the impact on high-frequency components. 2 CO1 C2
- 2(b) A 4x4 bits/pixel original image is given below. Apply histogram equalization by rounding the resulting pixels to the nearest integer. 5 CO2 C3

2	5	6	0
7	6	4	6
5	3	4	5
7	7	3	1

Fig. 1

- i. Apply histogram equalization to the image by rounding the resulting image pixels to integer.
- ii. Sketch the histograms of the original image and the histogram-equalized image.
- 2(c) Compare Histogram Equalization and Histogram Specification. In which scenarios does "Specification" outperform "Equalization"? Justify with an example. 3 CO2 C2
- OR
- 2(a) "Median filter technique is the best way to de-noise the image." Justify this statement, specifically mentioning its effect on "salt and pepper" noise. 2 CO1 C2
- 2(b) Sketch the histograms of the original image and the equalized image (From Fig. 1). Why does histogram equalization rarely produce a perfectly flat histogram? 5 CO2 C3
- 2(c) In histogram equalization, why must the transformation function behave strictly monotonically increasing? Use a diagram to support your answer. 3 CO2 C2
- 3(a) In the HSI color model, how are the "actual" color (Hue) and the "purity" (Saturation) of the color measured? Provide a diagram of the HSI color space. 3 CO2 C2
- 3(b) Define Thresholding. How can you convert a gray-scale image into a binary image using an average threshold value T ? Express this in functional form. 3 CO2 C2
- 3(c) Explain the difference between Convolution and Correlation in spatial filtering. Why is the filter rotated 180° in convolution? 2 CO1 C2
- 3(d) Given an image in RGB components, describe the mathematical steps to derive the corresponding CMY coordinates. 2 CO1 C2